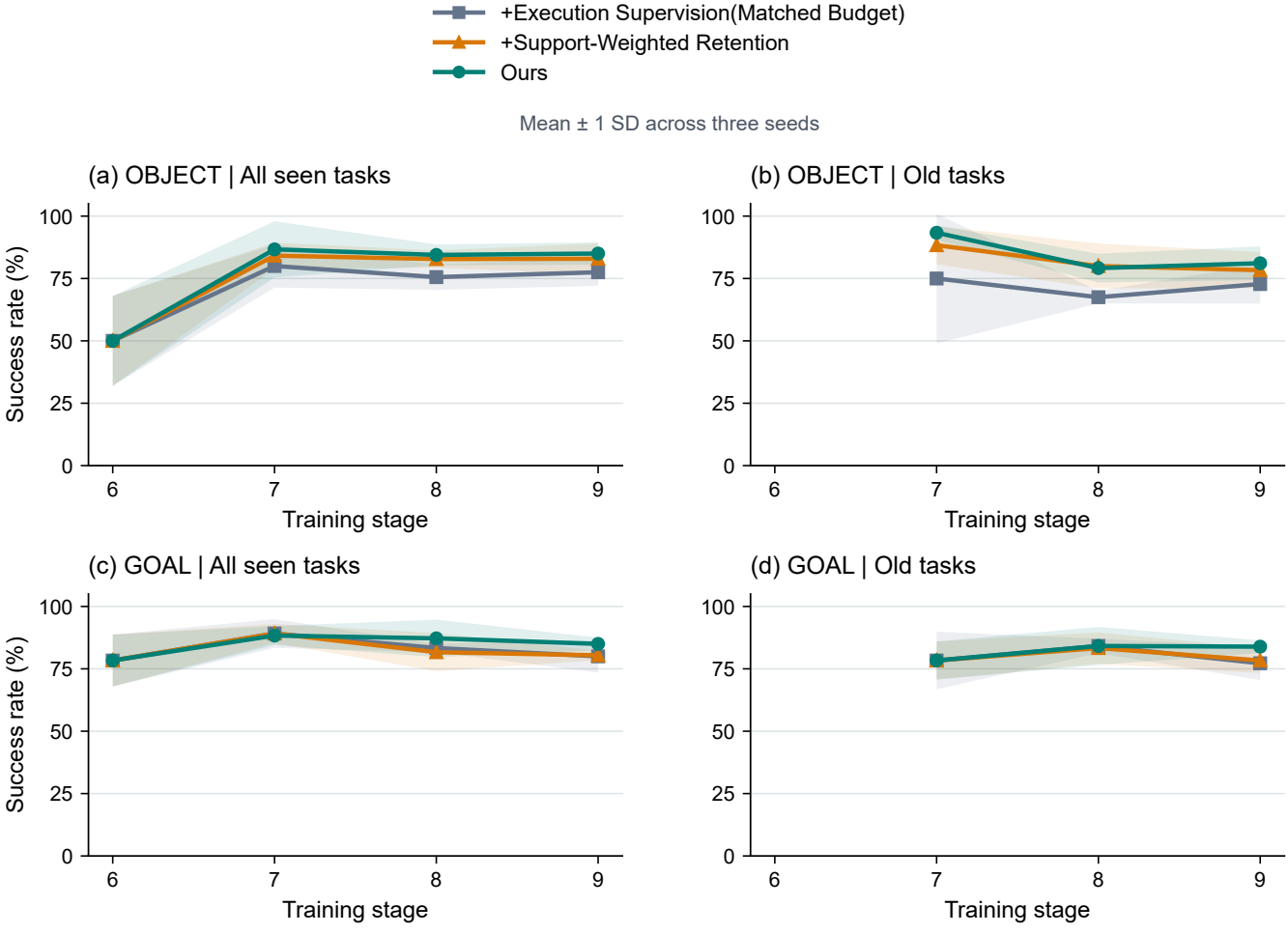
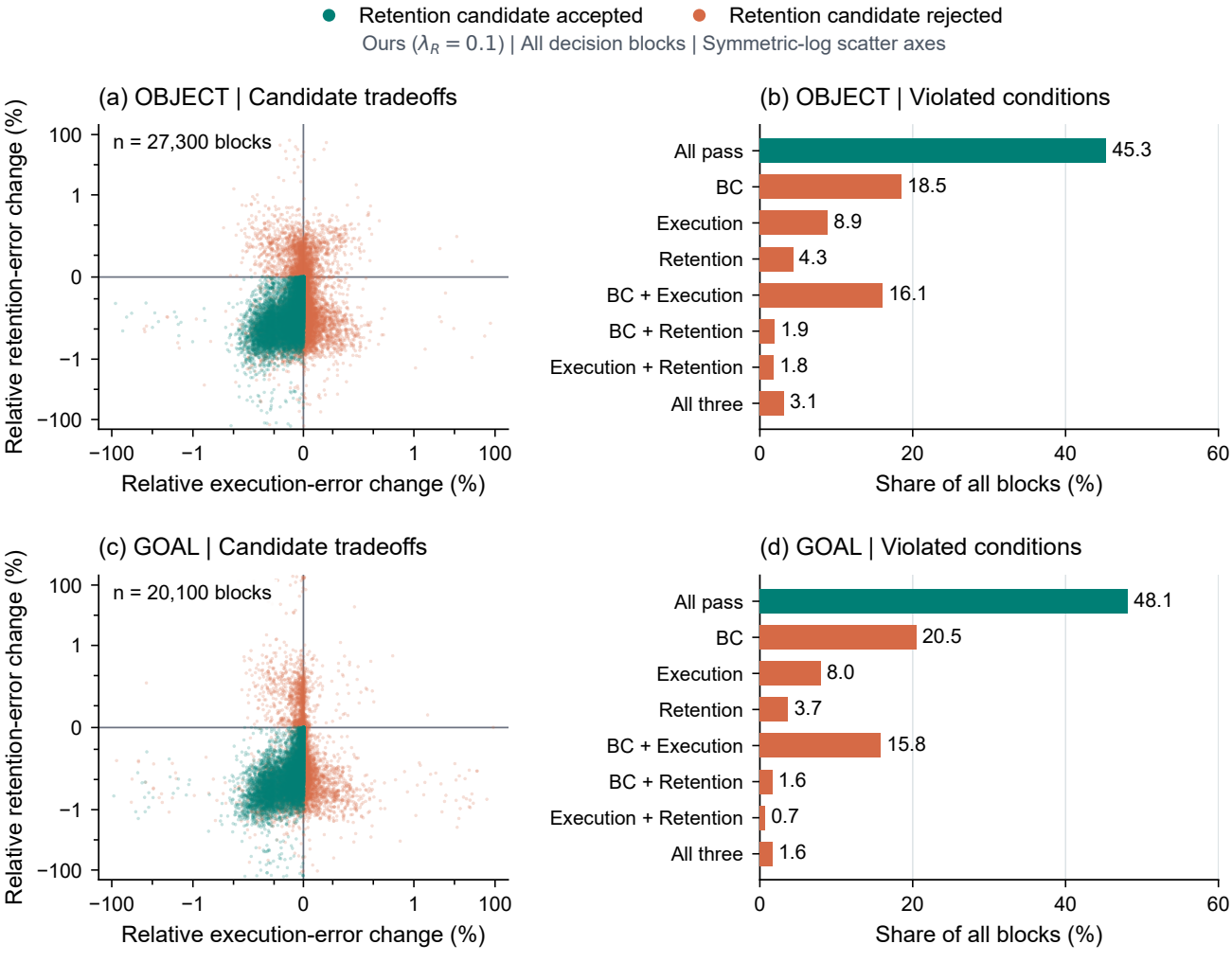
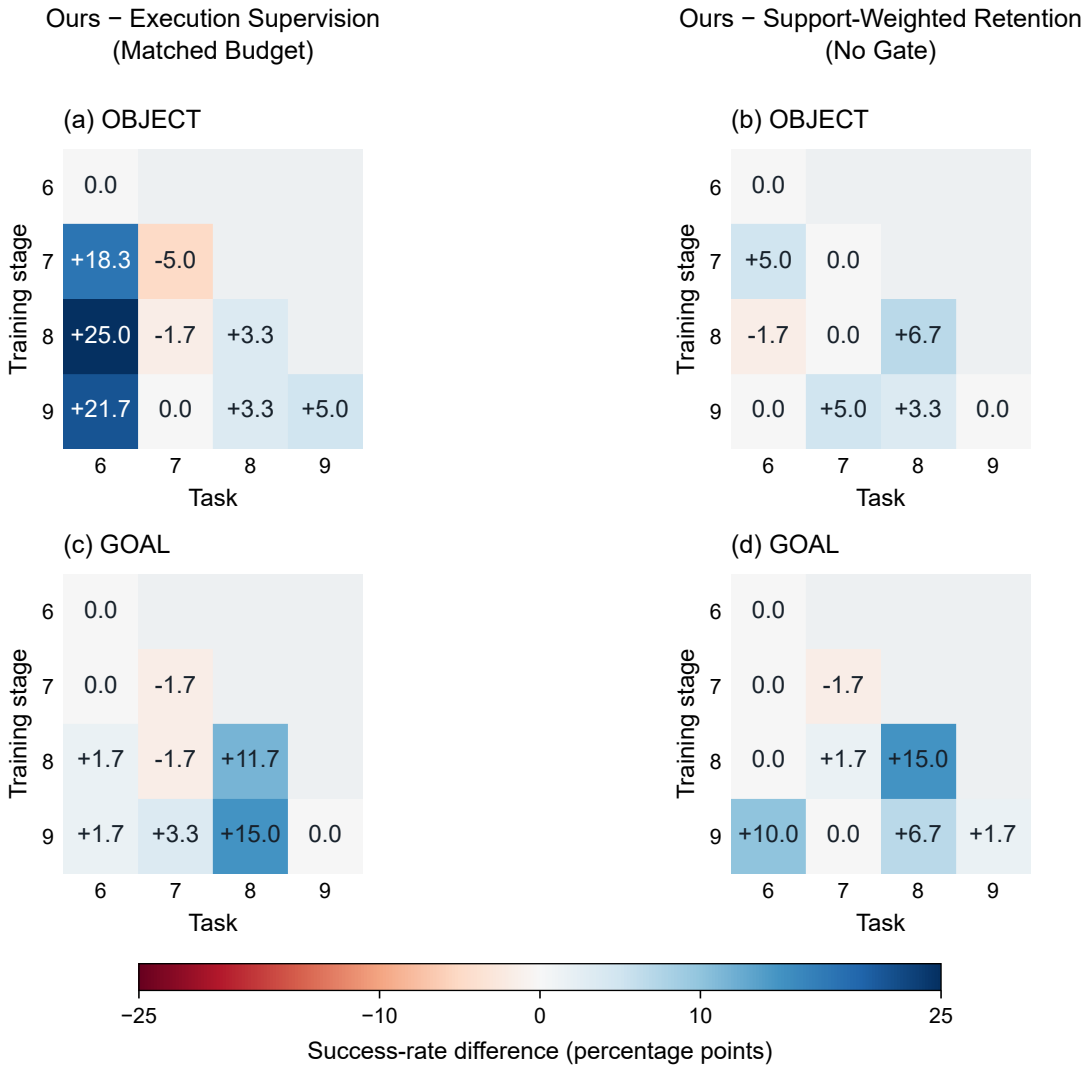

+Execution Supervision(Matched Budget)
+Support-Weighted Retention
Ours
Mean ± 1 SD across three seeds
(a) OBJECT | All seen tasks
(b) OBJECT | Old tasks
(c) GOAL | All seen tasks
(d) GOAL | Old tasks
Success rate (%)
Training stage
0
25
50
75
100
6
7
8
9


Retention candidate accepted
Retention candidate rejected
Ours ($\lambda_R = 0.1$) | All decision blocks | Symmetric-log scatter axes
(a) OBJECT | Candidate tradeoffs
n = 27,300 blocks
Relative retention-error change (%)
Relative execution-error change (%)
100
1
0
−1
−100
−100
−1
0
1
100
(b) OBJECT | Violated conditions
All pass
BC
Execution
Retention
BC + Execution
BC + Retention
Execution + Retention
All three
45.3
18.5
8.9
4.3
16.1
1.9
1.8
3.1
0
20
40
60
Share of all blocks (%)
(c) GOAL | Candidate tradeoffs
n = 20,100 blocks
Relative retention-error change (%)
Relative execution-error change (%)
(d) GOAL | Violated conditions
All pass
BC
Execution
Retention
BC + Execution
BC + Retention
Execution + Retention
All three
48.1
20.5
8.0
3.7
15.8
1.6
0.7
1.6
Share of all blocks (%)


Ours − Execution Supervision
(Matched Budget)
Ours − Support-Weighted Retention
(No Gate)
(a) OBJECT
(b) OBJECT
(c) GOAL
(d) GOAL
Training stage
Task
6
7
8
9
0.0
+18.3
-5.0
+25.0
-1.7
+3.3
+21.7
0.0
+3.3
+5.0
+5.0
0.0
+6.7
+5.0
+3.3
+11.7
+15.0
+10.0
−25
−10
0
10
25
Success-rate difference (percentage points)